

Design Resources & Project Documentation – Introduction to Python

Overview:

This document represents **design resources and project documentation** for an Introduction to Python lesson aimed at adult learners. It demonstrates the process of ideating, planning, and documenting a learning experience, including lesson objectives, storyboarding, learner personas, activities, assessments, and accessibility considerations. The focus is on **design process transparency, learner-centered planning, and scaffolded instructional design**—skills essential to a learning experience designer.

1. Lesson Outline – Python Basics

Lesson Title: Introduction to Python – Variables, Data Types, Tokens, Input/Print, Comments

Target Audience: Adult learners, beginner coders

Lesson Duration: 1.5 – 2 hours

Learning Objectives:

By the end of this lesson, learners will be able to:

- Understand Python syntax and structure
- Identify and use variables and data types
- Recognize Python tokens and reserved keywords
- Accept user input and display output using print()
- Write clear comments and understand their purpose
- Apply attributes to variables for simple operations

Lesson Activities & Assessment:

Activity	Description	Assessment
Introduction Video	Overview of Python, its uses, and syntax	None
Variables & Data Types	Interactive demo: learners create variables, assign values, check data types	Formative: short coding practice
Python Tokens & Keywords	Guided reading + examples	Quick quiz: identify valid tokens

Activity	Description	Assessment
Input & Print	Hands-on practice using input() and print()	Coding challenge: take user input and display output
Comments	Explain purpose of comments; learners write their own	Formative: check correct usage of comments
Mini Exercise	Combine variables, input, print, and comments	Submit small script for review

2. Storyboard – Mid-Stage Ideation

Screen #	Content	Interaction	Notes / Accessibility
1	Lesson introduction	Video + text	Explain Python and lesson objectives; captions, accessible font size
2	Variables	Interactive demo	Learners assign values, see output; scaffolded hints
3	Data Types	Drag-and-drop examples	Match variable types to values; alt text for visuals
4	Tokens & Keywords	Text + code examples	Learners highlight valid/invalid tokens; high contrast colors
5	Input & Print	Hands-on coding window	Learners type commands, see results; common error guidance
6	Comments	Interactive coding	Learners add comments; explain readability importance
7	Mini Exercise	Guided project	Combine all elements learned; hints available
8	Summary	Text + checklist	Recap key points, highlight takeaways; optional reflection question

Storyboard Purpose: Ensures each learning interaction is scaffolded and aligned with outcomes, supporting learner understanding and engagement.

3. Learner Personas

Persona	Prior Knowledge	Motivation	Goals	Notes
Alex, 32, Marketing Analyst	No prior coding; basic Excel	Upskill for data tasks	Write simple Python scripts for work	Prefers step-by-step instructions; hands-on practice
Priya, 27, Operations Coordinator	Some SQL knowledge	Career growth; interest in automation	Learn Python basics to automate reporting	Values real-world examples; needs short, clear exercises

Persona Use: These personas informed decisions about pacing, scaffolding, and accessibility in lesson design.

4. Design Overview & Accessibility Considerations

Screen	Title	Content	Interaction	Accessibility Notes
1	Welcome to Python Basics	Short video introducing Python, its uses, and lesson objectives	Play video; optional text summary	Captions, accessible font size, clear audio
2	Python Tokens & Keywords	Text + code snippet showing reserved keywords and tokens	Quiz: select valid tokens	Immediate feedback; alt text for snippet; high contrast colors
3	Python Data Types	Table showing int, float, str, bool + examples	Drag-and-drop activity	Alt text; real-life examples; high contrast
4	Understanding Variables	Text + example: <code>x = 5</code>	Learners type variable assignments	Hints for beginners; immediate feedback

Screen	Title	Content	Interaction	Accessibility Notes
5	Input & Print	Example: name = input("Enter your name: ") print("Hello, " + name)	Hands-on coding	Scaffolded guidance for common errors
6	Comments	Example: # This is a comment	Learners add comments	Explain readability and collaboration
7	Mini Exercise	Instructions to combine all elements	Submit code	Optional "Try Again" button; scaffolded feedback
8	Recap	Checklist of objectives	Learners tick off completed skills	Accessible font, icons, step-by-step scanning

Technology Use: Designed in Articulate 360 with interactive activities, quizzes, and drag-and-drop exercises. Focused on learner engagement, accessibility, and real-time feedback.

5. Reflection

This design documentation demonstrates my approach to **creating design resources and project documentation**. By developing lesson outlines, storyboards, and learner personas, I am able to **organize the learning experience**, scaffold instruction, and account for **learner needs and accessibility**. It also reflects thoughtful use of **instructional technology**, including Articulate 360, to foster interactivity and engagement. This document represents **original professional work** showcasing my capacity to plan, document, and communicate learning design effectively for adult learners.